CS 4400: Artificial Intelligence

Probability and Bayes' rule 2

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6/8. Independence

- Some variables are inherently unrelated
  - In P(Toothache,Catch,Cavity,Weather), weather does not affect nor is affected by other variables
- The notion is captured by independence
  - P(A,B) = P(A)P(B) : A and B are *independent*
- Example
  - Flipping 2 coins at once, is it independent?
  - $P(c_1=H,c_2=H) = P(c_1=H,c_2=T) = P(c_1=T,c_2=H) = P(c_1=T,c_2=T) = 0.25$
- More than two variables: P(A,B,C) = P(A)P(B)P(C)

Independence

- Independence can be expressed in other equivalent ways
  - Remember P(A,B) = P(A|B) P(B) = P(A) P(B|A) is always true
- A and B are independent iff
  - P(A,B) = P(A) P(B)
  - P(A) = P(A|B)
  - Prove that it's also equivalent to P(B) = P(B|A)
- Example
  - P(Toothache,Catch,Cavity,Weather) = P(Toothache,Catch,Cavity) P(Weather)
  - P(Toothache,Catch,Cavity|Weather) =P(Toothache,Catch,Cavity)
  - P(Weather|Toothache,Catch,Cavity) = P(Weather)

Conditional Independence

- Independence is a strong assumption
  - e.g., toothache and catch are not independent (they are each more likely if you have a cavity)
- However, variables may be independent GIVEN other variables
- Example:
  - $P(\text{toothache} \wedge \text{catch} \mid \text{cavity}) = P(\text{toothache} \mid \text{cavity}) P(\text{catch} \mid \text{cavity})$
    - Interpretation1: If you have a cavity, the probability that the probe catches in your tooth is not affected by your toothache
    - Interpretation 2: If you have a cavity, the probability of you having a toothache is not affected by whether the probe catches it or not

Conditional independence

- There are equivalent forms of conditional independence
- A and B are conditionally independent given C iff
  - P(A, B | C) = P(A | C) P(B | C)
  - P(A | B, C) = P(A | C)
  - P(B | A, C) = P(B | C)
  - Proof: ?
- Conditional independence is our most basic form of knowledge about stochastic environments

Cautions

- P(A,B) = P(A) P(B) means that it holds for all assignments of A and B
  - For example, if A and B are binary random variables, then
    - P(A=T,B=T) = P(A=T) P(B=T) (or $P(a \wedge b) = P(a)P(b)$)
    - P(A=T,B=F) = P(A=T) P(B=F) (or $P(a \wedge \sim b) = P(a)P(\sim b)$)
    - P(A=F,B=T) = P(A=F) P(B=T) (or $P(\sim a \wedge b) = P(\sim a)P(b)$)
    - P(A=F,B=F) = P(A=F) P(B=F) (or $P(\sim a \wedge \sim b) = P(\sim a)P(\sim b)$)
  - Is a much stronger claim than $P(a \wedge b) = P(a)\,P(b)$
- It can also be mixed: P(A, b) = P(A) P(b)
- The convention applies to all other equations

Combining evidence with conditional independence

- Suppose dentists knows

  $P(toothache|cavity), P(toothache|\sim cavity)$

  $P(catch|cavity), P(catch|\sim cavity)$

  $P(cavity)$

- and that

  $$P(toothache, catch|cavity) = P(toothache|cavity) \; x \; P(catch|cavity)$$

- What is the posterior $P(cavity|toothache, catch)$?

Combining evidence with conditional independence

- Follow the definitions

$$P(cavity|toothache,catch\ )$$

$$= \frac{P(cavity, toothache, catch)}{P(toothache, catch)}$$

$$= \frac{P(toothache, catch\ |cavity\ )P(cavity)}{P(toothache, catch)}$$

$$= \frac{P(toothache|cavity)P(catch|cavity)P(cavity)}{P(toothache, catch)}$$

Combining evidence with conditional independence

- Numerator: plug-in the values of
  $P(toothache|cavity), P(catch|cavity), P(cavity)$
- Denominator: a bit more computation required
- Direct approach:

$P(toothache, catch)$

$= {}_{Cavity} P(toothache, catch, Cavity)$

$= {}_{Cavity} P(toothache|Cavity)P(catch|Cavity)P(Cavity)$

$= {}_{Cavity} numerator$

Combining evidence with conditional independence

- Normalization approach: avoid computing the denominator

$$P(cavity \mid toothache, catch)$$

$$= \frac{P(toothache \mid cavity)\, P(catch \mid cavity)\, P(cavity)}{\sum_{Cavity} P(toothache \mid Cavity)\, P(catch \mid Cavity)\, P(Cavity)}$$

$$= \alpha P(toothache \mid cavity) P(catch \mid cavity) P(cavity)$$

Here α (= 1/denominator) is called a normalizing constant

Combining evidence with conditional independence

- How to compute α ?
- From the probability axioms:

  $P(cavity|toothache, catch) + P(\sim cavity|toothache, catch) = 1$
- Since

  $P(Cavity|toothache, catch) =$
  $\alpha P(toothache|Cavity)P(catch|Cavity)P(Cavity),$

  we have

  $\alpha P(toothache|cavity)P(catch|cavity)P(cavity) +$
  $\alpha P(toothache|\sim cavity)P(catch|\sim cavity)P(\sim cavity) = 1$
- Plug-in the probabilities, solve for α, and finally get $P(cavity|toothache, catch)$

Naïve Bayes (optional)

- This type of pattern is known as naïve Bayes:

$$P(Cause, E_1, \ldots, E_n) = P(Cause) \prod_i P(E_i|Cause)$$

Or equivalently:

$$P(Cause|E_1, \ldots, E_n) = \frac{P(Cause) \prod_i P(E_i|Cause)}{\sum_{Cause} P(Cause) \prod_i P(E_i|Cause)}$$

  - Naïve because it is often used as a simplifying assumption in places where the Effect variables are not actually conditionally independent
    - And yet it works surprisingly well in practice anyway

Naïve Bayes Example

- Bayesian spam filter
  - An e-mail may be a spam or ham (Spam=T or F)

$$P(Spam|F_1, \ldots, F_n) = \frac{P(Spam) \prod_i P(F_i|Spam)}{\sum_{Spam} P(Spam) \prod_i P(F_i|Spam)}$$

  - Here the "effect" terms are *features* from the underlying email
    - Words used in the email ("*Dear Sir/Ma'am, I'm a prince from Nigeria …*")
    - Sender information (blacklist)
    - Contents: images, attachments etc.
    - CC'ed to everybody

Naïve Bayes Example

$$P(Spam|F_1, \dots, F_n) = \frac{P(Spam)\ _i\ P(F_i|Spam)}{_{Spam}\ P(Spam)\ _i\ P(F_i|Spam)}$$

- Building a spam filter (=machine learning)
  - Collect statistics of the co-occurrence of features and true spam/ham labels from many emails
  - Compute scores $P(F_i|Spam)$ for all features

    (How often does a feature show up in spam vs ham?)
- Using a spam filter given a mail
  - Compute the posterior $P(Spam|F_1, \dots, F_n)$
  - If $P(Spam = T|F_1, \dots, F_n) \geq P(Spam = F|F_1, \dots, F_n)$, then it's a spam. Otherwise, it's a ham

Naïve Bayes Example

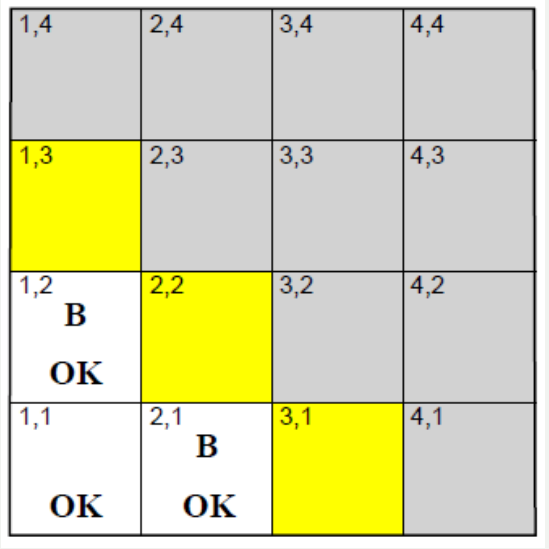
$$P(Spam|F_1, \ldots, F_n) = \frac{P(Spam) \prod_i P(F_i|Spam)}{\sum_{Spam} P(Spam) \prod_i P(F_i|Spam)}$$

$$= \alpha P(Spam) \prod_i P(F_i|Spam)$$

- Notice that we don't actually need the normalization term (=α), if we are only interested in deciding spam vs ham
- Still need to compute α, if we want the actual probability of a mail being a spam/ham

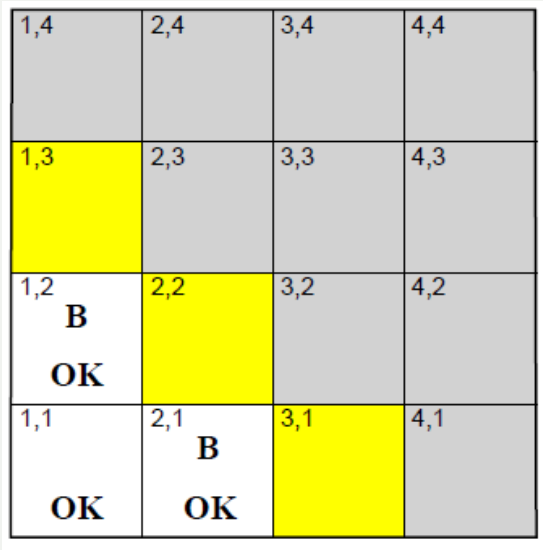
8/8. Probabilistic Wumpus world

- An agent has explored [1,1], [1,2], and [2,1]
- Observed breeze at [1,2] and [2,1]
- Breeze is felt adjacent to squares with pits
- Stochastic task environment:
  - Each square can independently have a pit with 0.8 probability (therefore no pit with probability 0.2)
- Query: which square is better, [1,3] or [2,2]?



8/8. Probabilistic Wumpus world

- Random variables
  - $P_{i,j}$ = true iff [i,j] contains a pit
  - $B_{i,j}$ = true iff [i,j] is breezy
- Consider variables $P_{1,1},\ldots, P_{4,4,}$ and $B_{1,1}$, $B_{1,2}$, $B_{2,1}$ only



Specifying the probability model

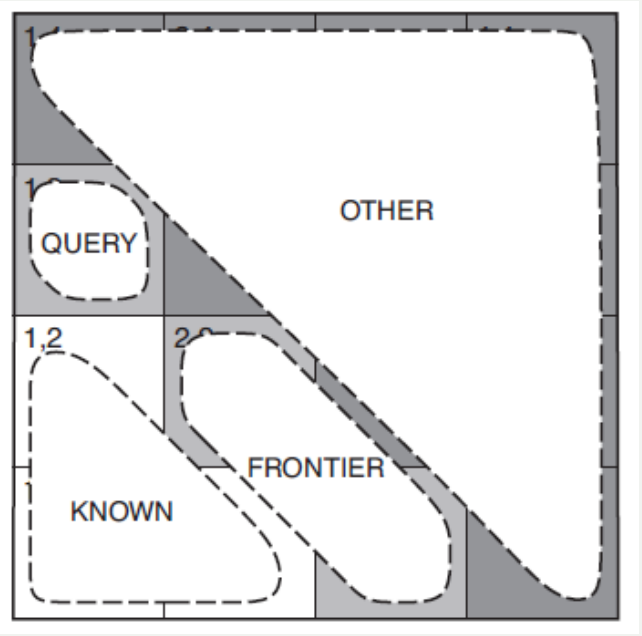
- Full joint distribution
  - $P(P_{1,1}, \ldots, P_{4,4}, B_{1,1}, B_{1,2}, B_{2,1})$
- Use product rule
  - $P(\text{joint}) = P(B_{1,1}, B_{1,2}, B_{2,1} \mid P_{1,1}, \ldots, P_{4,4})\, P(P_{1,1}, \ldots, P_{4,4})$
  - First term: 1 if pits are adjacent to breezes, 0 otherwise
  - Second term: pits are placed randomly, probability 0.2 per square:
  - $P(P_{1,1}, \ldots, P_{4,4}) = \prod_{i,j=1,1}^{4,4} P(P_{i,j})$
  - If there are *n* pits at fixed locations, then $P = 0.2^{n} \times 0.8^{16-n}$
  - What if we wrote $P(\text{joint}) = P(P_{1,1}, \ldots, P_{4,4} \mid B_{1,1}, B_{1,2}, B_{2,1}) \times P(B_{1,1}, B_{1,2}, B_{2,1})$?

Observation and query

- Define symbols for what we already know
  - known ≡ $\sim p_{1,1} \wedge \sim p_{1,2} \wedge \sim p_{2,1}$
  - b ≡ $\sim b_{1,1} \wedge b_{1,2} \wedge b_{2,1}$
- Query: $P(P_{1,3} \mid \text{known}, b)$
- How to proceed?
  - Rewrite the posterior as
  - $P(P_{1,3} \mid \text{known}, b) = \alpha \sum_{\text{Unknown}} P(P_{1,3}, \text{known}, \text{Unknown}, b)$
  - "Unknown" means all $P_{i,j}$'s except for $P_{1,3}$ and "known"
  - # of terms in the summation equals 2^(# of unknown squares)
- Grows exponentially with number of squares!

Using conditional independence

- Basic insight: $B_{i,j}$ (=breeze at [i,j]) is, given the presence/absence pits in adjacent squares, conditionally independent of other squares

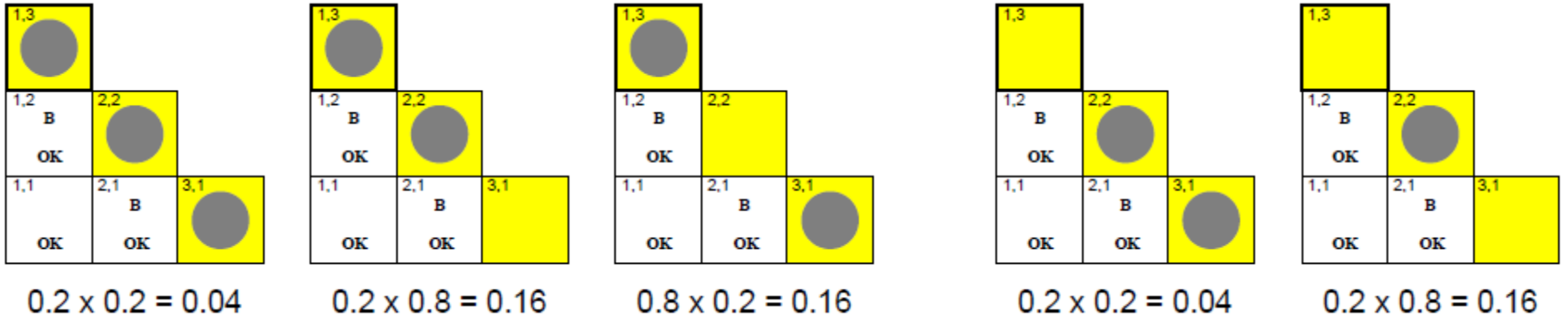


- We are only concerned with

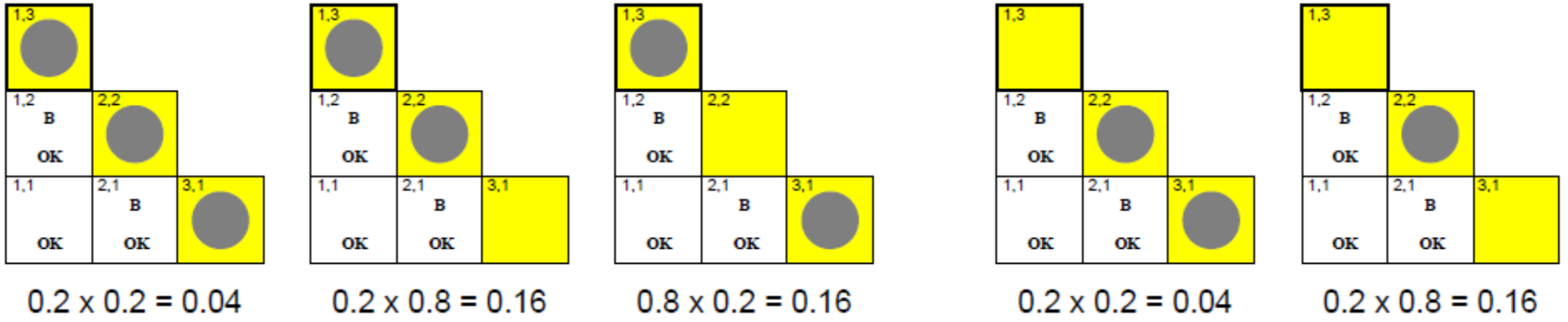
  $b \equiv \sim b_{1,1} \wedge b_{1,2} \wedge b_{2,1}$

- Group "Unknown" pits as "Frontier(or fringe)" or "Other"
- $P(b|P_{1,3},Known,Unknown) = P(b|P_{1,3},Known,Frontier)$
- Use this to answer the query!

$$
\begin{aligned}
\mathbf{P}(P_{1,3}|known, b) &= \alpha \sum_{unknown} \mathbf{P}(P_{1,3}, unknown, known, b) \\
&= \alpha \sum_{unknown} \mathbf{P}(b|P_{1,3}, known, unknown)\mathbf{P}(P_{1,3}, known, unknown) \\
&= \alpha \sum_{fringe} \sum_{other} \mathbf{P}(b|known, P_{1,3}, fringe, other)\mathbf{P}(P_{1,3}, known, fringe, other) \\
&= \alpha \sum_{fringe} \sum_{other} \mathbf{P}(b|known, P_{1,3}, fringe)\mathbf{P}(P_{1,3}, known, fringe, other) \\
&= \alpha \sum_{fringe} \mathbf{P}(b|known, P_{1,3}, fringe) \sum_{other} \mathbf{P}(P_{1,3}, known, fringe, other) \\
&= \alpha \sum_{fringe} \mathbf{P}(b|known, P_{1,3}, fringe) \sum_{other} \mathbf{P}(P_{1,3})P(known)P(fringe)P(other) \\
&= \alpha\, P(known)\mathbf{P}(P_{1,3}) \sum_{fringe} \mathbf{P}(b|known, P_{1,3}, fringe)P(fringe) \sum_{other} P(other) \\
&= \alpha' \mathbf{P}(P_{1,3}) \sum_{fringe} \mathbf{P}(b|known, P_{1,3}, fringe)P(fringe)
\end{aligned}
$$



- $P(p_{1,3} \mid known, b)$
- $= \alpha' P(p_{1,3}) \sum_f P(b \mid k, p_{1,3}, f) P(f)$
- $= \alpha' 0.2*(0.04+0.16+0.16) = \alpha' 0.072$
- $P(\sim p_{1,3} \mid known, b) = \alpha' P(\sim p_{1,3}) \sum_f P(b \mid k, \sim p_{1,3}, f) P(f)$
- $= \alpha' 0.8*(0.04+0.16) = \alpha' 0.16$



- From $P(p_{1,3} \mid known, b) + P(\sim p_{1,3} \mid known, b)$
  $= \alpha' 0.072 + \alpha' 0.16 = \alpha' 0.232 = 1$, we get
  $P(\sim p_{1,3} \mid known, b) = \alpha' 0.072 = 0.31$

- Similarly, $P(p_{2,2} \mid known, b) = 0.86$
- Agent will choose [1,3] over [2,2]